

Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents

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Abstract

An interactive agent is supposed to use what it is told. We instrument Turtle Soup, a lateral-thinking game of yes/no questions, with two process-level probes, a forced and scored checkpoint story every round and an association trajectory embedded from each round’s question, and run three Qwen tiers on 22 human-played puzzles. Accuracy is flat after round ten at every tier, and the round budget is not the constraint. The reason is visible in the process: the Oracle’s answer barely changes what the agent does next. Controlling for round position, the smallest tier’s next move is statistically independent of whether it was just confirmed or refuted (Δ stride +0.001, 95% CI $[-0.009, +0.011]$); the larger tiers do respond, but backwards, moving *further* after a confirmation than after a refutation (+0.028 $[+0.010, +0.046]$ and +0.009 $[+0.002, +0.017]$), the opposite of a search that eliminates hypotheses. The two failure patterns follow. Unable to be pushed out of its first basin, the small model *circles* (late-game stride 0.030 vs 0.082). Free to settle next to a confirmed neighbour, the largest tier *abandons*: it reaches the solution in 6 games and commits 1, and the story it ends with sits no further from the one that scored 1.00 than any two rounds of the same game sit from each other (0.851 vs 0.841), which is why the geometry describes these failures without predicting scores. One endpoint number scores all of this identically. We release the harness, both instruments, the puzzle set, and every per-game log.

1 Introduction

Here is a Turtle Soup puzzle. *A man’s body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story: a hot-air balloon was losing altitude; the passengers threw out their luggage, still could not stay up, and drew lots with matchsticks; the man who drew the short one was thrown out. The solver recovers this story by asking yes/no questions, one per round.

Outcome metrics can say whether an agent got there; on our set that story is flat, and the round budget is not the constraint (§5). What they cannot say is how the agent searched: one zero-scorer never leaves its first hypothesis, another finds the answer and lets it go. Instruments that read the process, and an environment audited enough to trust them, are this paper’s contribution. Turtle Soup suits the job: the behavior *is* text, the Oracle’s reply is a clean one-trit signal whose effect on the next move can be isolated, and puzzles are deliberately under-determined.

An interactive agent is supposed to *use* what it is told, so we ask first whether it does: does the answer just received change what the agent does next? Embedding each round’s question makes this measurable, and the answer is this paper’s spine, barely, and where it does, backwards. Three questions organize the trajectory analysis around it. **Q1**, does a failing agent *circle* a hypothesis basin, its stride (per-round semantic step size) shrinking toward zero, of which lexical novelty catches only the verbatim extreme? **Q2**, does it instead *drift*, striding normally while moving away from the anchor? **Q3**, do such trajectory features predict solve quality? The answers are yes, no, and no; and the checkpoint record reveals a second pattern the geometry did not hypothesize, *abandoning* (§5).

Contributions. (1) A measurement of **feedback response** in an interactive agent, stride conditioned on the answer received with round position controlled, absent at the smallest tier and inverted at the larger two. (2) Two failure patterns **derived from it rather than catalogued**, circling and abandoning, with per-game criteria and counts. (3) An honest boundary: the geometry describes these failures but cannot score them, because the gap between holding the solution and having lost it is no larger than ordinary within-game motion. (4) Released harness, both instruments, a 22-puzzle human-played set, and every per-game log.

Conditions of validity (an audited cross-family Oracle; one environment artifact removed) are stated in one sentence in §5 and specified in Appendix B; with a 95%-accurate Oracle they are no longer live problems for these results.

2 Related Work

Existing turtle-soup benchmarks, TurtleBench [1], SPLAT [2] (whose interactive player–judge protocol we adopt), and TurtleSoup-Bench [3], score verification accuracy, solve efficiency, or faithfulness to the intended story. TurtleSoup-Bench probes what agents ask next but grades it against the intended story; none of the three measures the search’s geometry, which is our delta. Circling and abandoning echo folk-known multi-turn failure modes (repetitive questioning; failing to commit or retain a correct answer); our contribution is to measure them, not to claim their novelty. MUTATE [4] shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow agent is a natural test intervention for our metrics. The instrument itself adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics. The Small World of Words norms [9] provide the human anchor that ours is not yet (§3). Judge-bias catalogues [5, 6] shape the scoring: the model judge is confined to the one subscore that objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 (the “surface”, the puzzle’s visible scenario) and asks one yes/no question per round. An **Oracle** holds the 汤底 (the hidden story) and answers 是/不是/与此无关 (yes, no, or irrelevant). The Questioner may commit a final story at any round, or is forced to at the budget. The open-source harness has pluggable model providers and logs every round, with per-game seeds.

3.2 Checkpoint and commitment instrumentation

Our first instrument does not wait for the end. Every round, after the Oracle answers, the Questioner must write its best complete story, and *that* checkpoint is scored; each point on an accuracy curve is therefore a full answer written with the evidence available at that round. Alongside it we record *commitment*: whether and when the agent ever volunteers a final story of its own accord. Together these turn an endpoint benchmark into a scored process.

3.3 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set therefore takes only puzzles with a public record of human play: 22 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102), with 97 annotated key clues. Provenance is enforced in code, with a verified family, a quarantined generated family, and tests that fail if the two mix.

Each puzzle carries an **under-determination** score: how far the surface alone leaves a solver from the solution, taken as the closest of twelve cold guesses written from the surface with no Oracle feedback. It ranges 0.173–0.577 (median 0.347) and enters §5 as a puzzle-level covariate; Appendix A gives its distribution and properties.

3.4 Outcome scoring: composite, not surface

Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**; Appendix B gives the detail. String matching alone cannot tell a wrong answer from a right one worded differently: we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Conversely, the clue half blocks “far = creative” gaming, because distance without validity scores zero.

3.5 Association-trajectory instrumentation

The second instrument reads the questions themselves. Each round, we extract the question’s keywords (jieba), embed them (bge-small-zh-v1.5), and average them to a unit round vector q_t ; distances are cosine, and anchors embed the puzzle’s surface, solution, and clue terms (specification in Appendix B). Two signals follow: the **stride** $s_t = d(q_t, q_{t-1})$ and the **anchor distance** $h_t = d(q_t, \mathcal{A})$, with Q1’s signature $\bar{s} \rightarrow 0$ and Q2’s a rising h_t ; the lexical `question_novelty` metric is the ablation baseline that Q1 predicts will miss synonym drift. One caveat up front: the anchor is a stated proxy, not a human association manifold, and its consequences are quantified with the results.

4 Experiment Designs

Designs. E1 (round curve) plays to 30 rounds under the checkpoint instrument (§3.2). E2 (round budget) imposes hard caps {5, 10, 15, 20, 25, 30} with a forced final answer only, so it doubles as a no-checkpoint control. E3 (association trajectory) computes $\{s_t\}$ and $\{h_t\}$ for every E1 game and answers Q1–Q3; the keyword extractor is fixed across models, so none is scored through its own vocabulary.

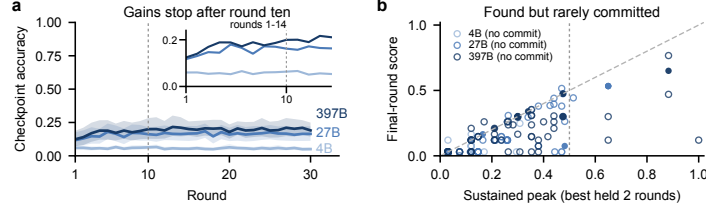


Figure 1: (a) Mean checkpoint accuracy by round on the full 0–1 range (bands, 95% cluster-bootstrap CIs over the 22 puzzles); dashed rule at round 10; inset magnifies the early rise. **(b)** Final score against sustained peak, one point per E1 game; filled markers are games whose final story was volunteered, open markers were never committed, and the dashed vertical rule marks solution-grade peaks (≥ 0.5 ; points may sit above the diagonal since the peak needs two consecutive rounds).

Feedback response, defined. For each round t we pair the Oracle’s answer (是 / 不是 / 与此无关) with s_{t+1} , the stride into the following question. A searcher that eliminates hypotheses should move *further* after 不是 than after 是. Confirmation arrives earlier in a game than refutation (mean round ~ 10 vs ~ 16) and early strides are larger, so the comparison is made inside (puzzle, round-bin) cells; CIs bootstrap over puzzles. Without that control the effect is manufactured at every tier, and both estimates are reported in Appendix B.

5 Results

All numbers come from one grid on the verified set: 22 puzzles $\times$ {Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B; hereafter 4B, 27B, 397B} $\times$ 3 seeds. **Conditions of validity, in one sentence:** the Oracle and logic judge are DeepSeek-V3.1, cross-family with the Questioners and audited at 95% on held-out probes, and one environment artifact (token exhaustion at the smallest tier) is removed in §5.1 before any behavioural comparison, with protocol, per-candidate audit numbers and settings in Appendix B. The three models differ in generation and architecture (the 397B is a mixture-of-experts with 17B active parameters), so we speak of model tiers, not scale; harness failures were zero, and token exhaustion is a game outcome reported below. Accuracy denotes the composite score rescaled to $[0, 1]$. E1 comprises 198 thirty-round games with a scored checkpoint every round; E2, 1,188 games across caps {5, 10, 15, 20, 25, 30}. Circling will emerge from the geometry (§5.3); abandoning from the checkpoint record (§5.1, §5.3).

5.1 Thirty rounds of evidence buy nothing after the tenth

Checkpoint accuracy rises only in the first ~ 10 rounds, and only for the larger tiers: 397B climbs from 0.124 (round 1) to 0.200 (round 10), then stays between 0.18 and 0.21; 27B rises 0.117 $\rightarrow$ 0.162 and plateaus near 0.16; 4B is flat at 0.05–0.06 throughout (Figure 1a). The round 10 to 30 change is $-0.009 / +0.003 / -0.013$ across tiers (puzzle-clustered 95% CIs spanning or hugging zero). E1 imposes no game token budget (the 50k budget governs E2 play) and no E1 game ended early, so the curves carry no survivorship.

Commitment: abandoning’s first evidence. 27B volunteers a final story in 11/66 games (mean round 11.1), 397B in 9/66 (14.2), and 4B in 2/66 (23.5); the smallest neither improves nor concludes, and the largest, as §5.3 quantifies, finds and does not commit.

Peaks, finals, and an honest null. Model tier raises what is found mid-game: the *sustained* peak (best score held over two consecutive checkpoints, robust to single-round rewording swings) averages 0.084 / 0.240 / 0.296 across tiers, while finals sit lower (Figure 1b). Retention (final over sustained peak; estimator and inclusion rules in Appendix B) is 0.69–0.76 at every tier, yet under a within-game permutation null (knowledge constant, only scoring jitter) it is expected at 0.79–0.84, and the observed values are not significantly below ($p \geq 0.09$). The quantitative case for “finding and losing” is therefore inconclusive; abandonment’s evidence is the commitment record above and §5.3’s cross-tabulation.

It is not the budget. End accuracy is insensitive to the round cap for the larger tiers (397B 0.19–0.21 and 27B 0.14–0.18 across all six caps). The smallest model’s raw curve declines monotonically (0.050 at cap 5 to 0.004 at cap 30), but this is an environment effect, not a behavioral one: games ending by token-budget exhaustion rise from 0/66 at cap 5 to 63/66 at cap 30 (175 of 396 overall, versus 2/396 for 27B and 0/396 for 397B), and conditioned on non-exhausted games 4B’s end accuracy does not decline (0.05–0.09 across caps). By our own audit principle, the raw decline measures the harness. This resolves an internal check: 4B’s E1 round-30 mean (0.05) and E2 cap-30 final (0.004) differ exactly by exhaustion; on non-exhausted games the protocols agree, as at the larger tiers.

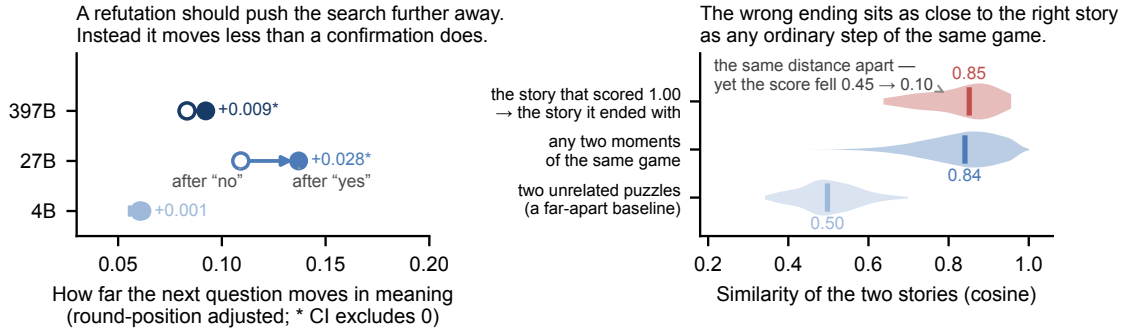


Figure 2: (a) How far the next question moves, after a refutation (open) and after a confirmation (filled), adjusted for round position; a search that eliminated hypotheses would move further after “no”. The smallest tier’s two values coincide: its next move does not depend on the answer. (b) In abandoning games, the story the model ends with sits as close to the story that scored 1.00 as any two moments of the same game sit to each other, while the score falls from 0.45 to 0.10. Losing a correct answer is therefore invisible to this geometry.

5.2 The answer barely changes what the agent does next

This is the paper’s central measurement. Pairing each answer with the stride into the following question, inside (puzzle, round-bin) cells so that the earlier arrival of 是 cannot manufacture the effect (Figure 2a): 4B +0.001 [−0.009, +0.011], no detected response; 27B +0.028 [+0.010, +0.046] and 397B +0.009 [+0.002, +0.017], both responding with the sign inverted. Two distinct pathologies, and neither is the search a rational agent would run.

The smallest tier does not hear the answer. Its next move is statistically independent of whether it was just confirmed or refuted, so nothing in the feedback channel can push it out of the hypothesis it started with. That is what circling is, now with a cause rather than a description.

The larger tiers hear it backwards. A refutation eliminates a hypothesis and should force a jump; a confirmation should invite a refinement. Both larger tiers do the reverse, and since 68–74% of answers are 不是, most rounds are spent not moving.

5.3 Two patterns: circling and abandoning (E3)

With the environment artifact removed, the differences are behavioral. **Circling (Q1, confirmed):** the geometry carries the signature the feedback measurement predicts. 4B’s mean stride is 0.049 [0.040, 0.060] against 0.114 (27B) and 0.088 (397B), and its late-game stride 0.030 [0.021, 0.039] against 27B’s 0.082 [0.065, 0.098]. Not a malformed-question artifact: 17% of 4B’s questions are truncated or non-interrogative (~1% elsewhere), yet excluding them barely moves either statistic (0.049 → 0.056; 0.030 → 0.033).

Abandoning (unhypothesized; from the checkpoint record): call a sustained peak ≥ 0.5 (half the scale) *solution-grade*. 4B reaches none; 397B reaches six, commits one, and ends four below half peak (ratios 0.12–0.87); 27B reaches four, commits three, keeping ~ 0.83 . Abandoning concentrates in the largest tier; Figure 4 shows one game whole. Under per-game criteria (circling: late stride < 0.04 , no commit, no peak ≥ 0.3 ; abandoning: peak ≥ 0.3 ending below half), the tiers swap dominant failures (4B, 39 vs 1 of 66; 397B, 6 vs 12), no game meeting both.

Abandoning is a sub-semantic event, and that is why the geometry cannot score it. In abandoning games the final story sits at cosine 0.851 from the peak story that scored up to 1.00, while any two checkpoints of the same game sit at 0.841 and two stories from different puzzles at 0.498 (Figure 2b). The move that loses the answer is indistinguishable, in this space, from ordinary within-game motion, though the score falls by 0.30 on average. This is a property of the phenomenon, not a defect of the instrument: a fatal substitution smaller than the encoder’s resolution cannot be predicted from that encoder.

Drifting (Q2, rejected): slopes cluster near zero at every tier. In language space, lexical novelty catches only 4B’s verbatim re-asks and cannot see abandoning at all.

Predictivity (Q3, not supported): within tiers no stride or drift effect is detected, every coefficient bounded within ± 0.035 per SD (Appendix B). Effects are bounded small, not excluded: the geometry says *how* an agent fails, not how well it will score.

6 Limitations and Conclusion

Limitations. The feedback measurement reads *how far* the next question moves, not *what* it asks; a content-level test is the natural next study, and round position is controlled only by binning. Truncated chain-of-thought under a tight token budget masquerades as an agent that cannot form a question (§5.1), and the Oracle’s ~95% audited accuracy implies roughly one wrong answer per 30-round game. The anchor is not yet human-derived (§3.5), and §5.3 shows the encoder’s resolution is itself a limit on what any embedding-based instrument can score. One DeepSeek model serves as both Oracle and judge, and the tiers confound generation and architecture, so tier effects are model comparisons, not scaling laws. The source site is public and memorisation probes are pending. Puzzles involve death and dark themes; content-flagged.

Conclusion. The flat curve is what happens when evidence does not move the search: one tier cannot be moved at all and circles, two are moved the wrong way and abandon. They need opposite fixes; the score cannot tell.

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A The puzzle set

Where the puzzles come from. A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable; an agent scoring zero on it teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup community: 22 puzzles people have played and solved, each keeping its source record. Every item was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

How hard are they? Each puzzle carries an **under-determination** score: give a model the surface alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to the real solution. The index is one minus that best guess, and is large when the surface leaves a solver far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

The median is 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577: single-peaked with a thin hard tail. Easiest are the 海龟汤 story itself (0.17) and the desert matchstick (0.19). Hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Two properties make this usable as a covariate. It describes the puzzle, not any agent’s performance on it. It is also independent of rounds used; a difficulty defined as “how many rounds this takes” would make an accuracy-versus-rounds plot confirm itself. Nor is it a proxy for surface length ($r = -0.22$, n.s.), so length dependence was not detected at $n = 22$ (a weak test). It is one measure rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

A caveat we report either way. The two puzzles scoring easiest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure reads familiarity as much as difficulty. A memorisation probe (asking for the solution with no surface given) separates the two and belongs in any use of this measure.

B Measuring agent performance

What counts as a correct answer. Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and reproducible. The logic judge rates only whether cause $\rightarrow$ mechanism $\rightarrow$ outcome matches and is told explicitly to ignore wording. Because single ratings are unstable on borderline answers, ratings are sampled and averaged (two samples per rating in the reported grid). The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be. Below that floor only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above the floor.

Does the logic half earn its thirty points? On the full grid, yes: each model earns a several-fold larger share of the logic half than of the clue half, and the halves correlate only weakly per game (Spearman $\rho = 0.30/0.35/0.06$ for 397B / 27B / 4B over 396 E2 games each). The 70/30 weighting itself is a design choice; every conclusion in §5 is readable from the two halves separately, though the absolute plateau level (~ 0.2) does depend on the weights. The logic judge rates the causal-logic half on its 30-point scale and is sampled twice per rating at temperature 0.2; the two samples agree exactly in 86% of ratings and differ by under one point of 30 on average.

Feedback response: definition, confound, and both estimates. For each round t the Oracle’s answer is paired with s_{t+1} , the stride into the next question; 与此无关 rounds are excluded. Confirmation arrives earlier than refutation (mean round 12.1 / 9.7 / 10.9 for 是 against 15.6 / 15.6 / 15.9 for 不是) and early strides are larger, so an uncontrolled comparison is confounded by position. Pairs are formed inside (puzzle, round-bin) cells with bins {1–5, 6–10, 11–15, 16–20, 21–25, 26–29}, and 95% CIs bootstrap over puzzles (4,000 resamples). Uncontrolled, all three tiers appear to respond (+0.024 [+0.004, +0.044]; +0.043 [+0.015, +0.070]; +0.034 [+0.023, +0.046]); controlled, the smallest does not (+0.001 [−0.009, +0.011]; +0.028 [+0.010, +0.046]; +0.009 [+0.002, +0.017]). The gap between the two is the point.

The semantic scale of abandoning. Within abandoning games (sustained peak ≥ 0.3 , final below half of it; $n = 34$), the final checkpoint story sits at cosine 0.851 (median) from the peak story, while any two checkpoints of the same game sit at 0.841 (median over all 77,906 within-game pairs from 60 games) and two stories from different puzzles at 0.498 (231 pairs). The score gap across that same pair is $0.447 \rightarrow 0.099$.

The round-1 prior. The round-1 checkpoint precedes the first Oracle answer and is therefore a cold guess. It holds 49% / 33% / 26% (median, by tier) of the game’s eventual best score, and correlates with it at $\rho = +0.37 / +0.57 / +0.47$ (permutation $p = 0.09 / 0.003 / 0.03$, $n = 22$ puzzles with seeds averaged). Under-determination, itself measured from twelve cold guesses by a model cross-family with every Questioner, predicts round-1 quality at $\rho = -0.34 / -0.54 / -0.47$ ($p = 0.12 / 0.010 / 0.029$) but not the final score.

Behavioural indicators against puzzle structure. Regressing each per-trace indicator on under-determination, seeds averaged, 5,000 permutations: stride, late stride, drift (both anchors), mean anchor distance and rounds played are all n.s. at every tier ($|\rho| \leq 0.32$, $p > 0.10$). Commit rate is negative at all three ($-0.32 / -0.25 / -0.43$) but pooled $p = 0.13$, and 1 of 24 tests reaching $p < 0.05$ is the chance expectation, so we report it as unresolved rather than as an effect.

The two score halves dissociate (a check on the scoring instrument, moved here from the body). Every model earns proportionally far more of the judged causal-logic half than the clue-recall half: 397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, and 4B 2.9 against 0.2, and the halves correlate only weakly per game (Spearman $\rho = 0.30 / 0.35 / 0.06$). Models reach the right causal shape well before, and often without, landing the annotated clue content.

Retention estimator and permutation null. Retention is the mean per-game final-to-sustained-peak ratio over games with a nonzero sustained peak; the final is the last played checkpoint, and excluding early-committing games changes each tier’s value by ≤ 0.03 (bootstrap 95% CIs $\pm \approx 0.08$). For the null: unit, the game. For each of 2,000 permutations, every game’s checkpoint sequence is shuffled within the game (knowledge-constant, jitter-only null), the sustained peak and the final (last element) are recomputed on the shuffled sequence, and retention is the mean over that tier’s games with a nonzero permuted peak. The reported p is one-sided per tier, the fraction of permutations whose null retention is at or below the observed value. Late-game stride means the second half of a game’s strides.

Mixed-model coefficients. Best checkpoint accuracy per game; Gaussian LMM, puzzle random intercept, tier fixed effect, ML fits, likelihood-ratio tests against the covariate-only model. Stride $\beta = -0.010$ per SD, 95% CI $[-0.035, 0.014]$, $p = 0.42$; drift $\beta = -0.011$, $[-0.035, 0.014]$, $p = 0.39$; stride $\times$ tier $p = 0.20$; surface-anchor drift $\beta = -0.020$, $[-0.044, 0.003]$, $p = 0.097$.

Environment settings and Oracle audit. Every model call uses max_tokens 2048 (thinking disabled, temperature 0.2) with a per-game budget of 50,000 tokens; exhausting the game budget forces a final answer and is recorded as the game outcome token_budget. The Oracle audit probes a candidate with 25 hand-crafted (question, expected-answer) pairs over three puzzles, 21 with a definite yes/no answer and 4 expecting “irrelevant”; the pass bar is 90% on the yes/no items. The deployed DeepSeek-V3.1 scored 20/21 (95%) on yes/no and 3/4 exact on “irrelevant”. Candidates measured in the pilot phase: a 4B Oracle with thinking disabled, 30% (rejected; this is the configuration behind abstract finding 4, where raising Oracle accuracy from 30% to 100% left the 4B Questioner’s score unchanged); the same model with thinking enabled, 70% (rejected); a same-family flash model, 90% (excluded for sharing the Questioners’ family); GLM-5.3-flash, 100% (used in earlier tooling).

Trajectory and difficulty pipeline. Keywords are extracted with jieba after stripping Latin characters and digits; each keyword is embedded with bge-small-zh-v1.5 (normalized), and the round vector is the re-normalized mean of its keyword embeddings. All distances are cosine. The solution-aware anchor embeds the puzzle’s surface, solution, and clue terms; the surface-only anchor embeds surface terms alone. The under-determination measure prompts GLM-5.3-flash (cross-family with all Questioners) for twelve cold guesses from the surface and scores closeness as embedding similarity between guess and solution under the same encoder, not with the composite score, so the covariate is not coupled to the outcome apparatus.

Artifacts. Harness, puzzle set with per-item source records, per-game logs (including every checkpoint story), and the analysis scripts behind each reported number are released at github.com/RainyFields/Project_Turtle_Soup. Puzzle surfaces and solutions originate from a public community repertoire and are redistributed with source attribution and a content flag.

Where the accuracy curve comes from. E1 does not score once at the end. Every round, after the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored. Each point on the accuracy curve is therefore a full answer written with the evidence available at that round.

One game, worked. Puzzle refsoup_021 (surface: a stormy night, a run-down temple, two photographs on the

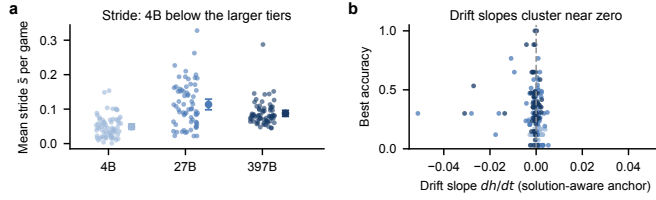


Figure 3: Per-trace geometry ($n = 198$). (a) Per-game mean stride by tier (dots; bars, tier mean $\pm$ 95% CI): 4B sits below the larger tiers. (b) Drift slope against best accuracy, clustered near zero at every tier: Q2’s absent signature.

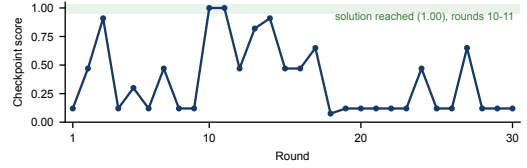


Figure 4: One abandoning game (refsoup_021, 397B, seed 2): the round 10–11 checkpoint stories are the solution (band), never volunteered, abandoned by round 19. Bilingual rows in Appendix B.

wall that seem to stare; by morning the narrator sees only two windows; solution: 墙上根本没有照片，是有人贴着窗户往里看，“there were never photos on the wall; people were pressing against the windows, looking in”), played by Qwen3.5-397B, seed 2. Table 1 shows seven of thirty rounds; Chinese cells are verbatim from the log with our translations, stories abridged to the clause carrying the hypothesis. Rounds 22–30, eight lexically distinct re-asks of the same flooding hypothesis (e.g. round 25, whether the narrator woke floating upside down; round 28, whether rising water reached eye level), are in the released logs.

Rd	Question (abridged)	Ans.	Committed story (abridged)	score (clue+logic)
1	照片里的是人吗? <i>Are the ones in the photos people?</i>	是 yes	那两张“照片”……是两个被钉死或封在墙洞里的人 <i>The two “photos” are two people nailed or sealed into wall niches.</i>	0.12 (0+12)
3	主角第二天醒来时，还在那座寺庙里吗? <i>On waking the next day, is he still in the temple?</i>	是 yes	所谓的“两张照片”其实是两扇窗户……误看成了照片 <i>The “two photos” are really two windows, mistaken for photos in the dim light.</i>	0.91 (70+21)
4	醒来看到的“两扇窗户”就是昨晚的“两张照片”吗? <i>Are the “two windows” the same objects as last night’s “photos”?</i>	是 yes	“照片”实际上是两个被挖空的人形窗口……恐惧感源于黑夜和错觉 <i>The “photos” were hollowed-out window openings; the fear came from darkness and illusion.</i>	0.12 (0+12)
10	“两扇窗户”是因为相框玻璃破碎脱落吗? <i>Did frame glass break and fall, leaving open holes?</i>	不是 no	那根本不是照片，而是……贴在窗玻璃外侧的人脸 <i>They were never photos, but faces pressed against the outside of the windows.</i>	1.00 (70+30)
11	是因为照片被取走、露出了后面的窗户吗? <i>Were the photos removed, revealing windows behind?</i>	不是 no	昨晚“盯着他的人像”，其实是窗外的人 <i>The staring figures were people outside the windows.</i>	1.00 (70+30)
19	人像是躲在照片后面的真人吗? <i>Were real people hiding behind the photos?</i>	不是 no	窗户洞口被画着人像的布帘封住……雷雨导致脱落 <i>Window openings sealed by painted cloth covers; the storm tore them off.</i>	0.12 (0+12)
30	是因为积水使主角身体位置升高吗? <i>Did rising water lift the narrator higher?</i>	不是 no	两个被纸糊住的窗户洞口……雷雨导致纸张脱落 <i>Papered-over window openings; the storm stripped the paper away.</i>	0.12 (0+12)

Table 1: Seven rounds of one E1 game (refsoup_021, Qwen3.5-397B, seed 2): the Questioner’s question, the Oracle’s answer, the story the agent was forced to commit that round, and its composite score. Italics are our translations.

Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowledge the agent never acts on.** The round-10 committed story *is* the solution, synthesized from earlier answers on a round whose own question was answered 不是. Yet the model never volunteers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Stalling in fresh words happens at the largest scale too.** Rounds 22–30 are eight lexically distinct re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **The curve’s round-to-round jitter is a scoring fact, not a knowledge fact.** The same understanding re-worded (rounds 3 $\rightarrow$ 4) swings the clue half 70 $\rightarrow$ 0. The mean curves of \$5 average this out, which is why single-game curves are not the unit of analysis.